MIL-C-62123A(AT)
22 June 1987
SUPERSEDING
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18 February 1970

MILITARY SPECIFICATION

CEMENT, TREAD, RUBBER

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers commercially manufactured cements containing resins which are used by the tire retreading industry for applying hot freshly extruded tread rubber compound to buffed tire casings upon which such resin base cements have been sprayed (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2640

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SPECIFICATIONS
FEDERAL

ZZ-T-416	- Tire, Pneumatic: Retread and Repair Materials.
ZZ-T-441	- Tire, Pneumatic: Retreaded and Repaired.

STANDARDS
MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-45662	- Calibration Systems Requirements.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D1084	- Viscosity of Adhesives, Test Method for.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA, 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

"3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish tread rubber cement which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representive of the cement to be furnished to the Government. All subsequent tread cement delivered to the Government shall conform to these samples in all their pertinent physical and performance attributes".

3.2 Material suitability. The cement to be furnished shall be similar to resin based cements developed for, and used by, the tire retread industry for production of satisfactory quality retread tires built with hot, extruded tread compound as specified in ZZ-T-416. The finished product shall conform to the requirements of ZZ-T-441 when standard shop practice is used for retreading (see 4.8.1).

3.3 Material content.

3.3.1 Rubber content. The cement shall contain natural, synthetic natural (cis 1-4 polyisoprene) or styrene-butadiene rubbers in any combination, but shall not contain reclaimed rubber (see 4.8.1).

3.3.2 Resin content. The cement shall contain resins dissolved in organic solvents such that the hot extruded tread rubber, when applied at a temperature of 200°F, will be held firmly to a buffed, cemented tire carcass without loosening, sagging, or drooping during and after the period in which the tread stock cools (see 4.8.1).

3.4 Adherence capability. The buffed cemented tire shall retain the capability of accepting and adhering to the hot extruded tread rubber for at least 5 hours after the cement has been applied (see 4.8.2).

3.5 Adhesion. When tested as specified in paragraph 4.8.3, a cured adhesion of 120 pounds per inch width shall be attained.

3.6 Solubility of resin cements. The addition of rubber solvent, Type IV, Class D, of ZZ-T-416 shall not cause galling of cement. The addition of rubber solvent shall not cause precipitation or settling in the cement so that it cannot be re-dispersed by stirring (see 4.8.2).

3.7 Viscosity. When tested as specified in 4.8.4, the viscosity of the resin cement shall be suitable for spray application as removed from the container. Viscosity shall range between 20-65 centipoises at 23 degrees Celsius (°C) [73.4 degrees Fahrenheit (°F)].

3.8 Total solids. When tested as specified in 4.8.5, the solids content of the cement shall not be less than 7 percent (%) by weight.

3.9 Storage stability. The cement shall be usable for 1 year from date of manufacture when stored at temperatures not exceeding 75°F (see 4.8.1).

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3.10 Workmanship. Workmanship shall be such as to produce cement which will be free of foreign material, cracks, cuts, blisters, or folds. Mixing of compound formulation will be complete and of the best commercial practice to adequately disperse all ingredients of the formula (see 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10% of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection:

a. First article inspection (see 4.4).

b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Tests (see 4.5.3).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

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a. Air temperature 73 ± 18°C
b. Barometric pressure 28.5 + 2 inches mercury (Hg)
- 3
c. Relative humidity 50 ± 30 percent

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select five gallons of cement produced under the production contract for first article inspection. First article samples shall be inspected as specified in table I. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply tread cement that is fully representative of cement inspected as a first article sample. Any changes or deviation of the production cement from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 First article inspection failure. Test item deficiencies during, or as a result of, the first article test shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, the first article test shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government shall not accept products until first article testing is completed to the satisfaction of the Government.

TABLE I. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance	
				Examination	Tests
Material suitability	3.3.1, 3.3.2 and 3.9	4.8.1	X		
Defects	3.4, 3.6, and 3.10	4.8.2	X	X	
Adhesion	3.5	4.8.3	X		X
Viscosity	3.7	4.8.4	X		X
Total solids	3.8	4.8.5	X		X

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4.5 Quality conformance inspections.

4.5.1 Lot formation. Unless otherwise specified (see 6.2), a lot shall consist of all cement from an identifiable production period (not to exceed 1 week), from one manufacturer and submitted at one time for acceptance. Each 50 gallons of cement shall be considered as one unit for sampling purposes. Five gallon specimens from the units of quantity to be sampled shall be selected for quality conformance inspection.

4.5.1.1 Optional sampling method. At the option of the Government, samples may be selected from closed (ready for shipment) packages. Such packages shall then be reweighed and resealed, provided the original weight has not been reduced by more than 20% and provided the package has been marked with the new weight and the word "SAMPLED".

4.5.2 Quality conformance examination.

4.5.2.1. Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2.2 Acceptable quality level. Each sample selected in accordance with 4.5.2.1 shall be examined for conformance to the following acceptable quality level (AQL), on the basis of percent defective:

Classification	AQL
Major	1.0

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table II.

TABLE II. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Adherence, Cement or application improper (see 3.4).	Visual
102	Solubility, galling, precipitation, or settling of cement (see 3.6).	Visual and tactile
103	Workmanship, nonconformance (see 3.10).	Visual

4.5.3 Quality conformance tests. Samples selected in accordance with 4.5.1 or 4.5.1.1 shall be subjected to the tests specified in table II, using an AQL of 4.0% defective.

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4.8 Methods of inspection.

4.8.1 Materials. Conformance to 3.3.1, 3.3.2 and 3.9 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports and rating data.

4.8.2 Defects. Conformance to 3.4, 3.6 and 3.10 shall be determined by examination for the defects listed in table II. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.8.3 Cement adhesion. To determine conformance to 3.5, two Government control specimens conforming to ZZ-T-416, each 6 inch square, backed on one side with fabric treated for adhesion, shall be used for this test. One specimen with total thickness of 0.125 inch (in) shall have been cured for 30 minutes at a temperature of 280°F. The other specimen of 0.125 thickness shall be uncured. Cured specimens shall have mold sheen on the side away from the fabric removed by stone buffing to a velvet finish prior to the application of cement. Buffed specimens shall be wiped before and after buffing with a cloth dampened with rubber solvent to insure clean surfaces. Cement shall be applied to matching 3-in portions of the two specimens. A material separating heat resisting film shall be placed between the two uncoated portions of the specimens and the specimens shall then be placed in a mold and cured for 45 minutes at 280°F. The finished block shall be 0.250 $\pm$ 0.025 inch thick. Each vulcanized specimen shall be cut into five strips each 1 inch wide and 6 inches long, and each with a 3 inch portion which has remained unbonded. Testing shall be conducted no sooner than 16 hours after cure and after the test specimens have been cooled to a temperature of 73° $\pm$ 2°F. These tests shall be run at the same temperature. Each free end of the unbonded portion of the test specimen shall be clamped into the testing machine (see 4.8.3.1). When the one portion is attached to the recording head, care shall be taken to adjust it symmetrically in order that tension shall be distributed uniformly. The autographic mechanism and chart shall be adjusted to zero and the machine started. The separation process shall be continued for a sufficient distance to indicate the adhesion value. The adhesion strength shall be determined by drawing on the autographic chart the best averaged load line which will accommodate the recorded curve. The load so indicated, expressed in pounds per inch in width, shall be reported as the adhesion strength of the vulcanizing cement under test. For each series of tests, the arithmetic mean of all the values obtained shall be calculated and reported as the "average value".

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4.8.3.1 Apparatus. The apparatus shall consist of a testing machine, preferably of the inclination-balance or pendulum type, which shall fulfill the following requirements:

a. The applied tension as measured and recorded shall be accurate within plus or minus 1%.
b. Specimens shall be held in the testing machine by clamps which grip firmly and prevent slipping at all times.
c. The machine shall be autographic, giving a chart having the inches of separation as one axis and applied tension as the other axis of coordinates.
d. The machine shall be of such capacity that the maximum applied tension during test shall not exceed 85% nor be less than 15% of the rated capacity.
e. Rate of jaw separation in testing shall be 2 inches per minute.

4.8.4 Viscosity. To determine conformance to paragraph 3.7, the test shall be conducted in accordance with method B of ASTM D1084.

4.8.5 Total solids.

4.8.5.1 Procedure. To determine conformance to 3.8, approximately 10 grams of the sample shall be poured into a low-form weighing bottle, covered and weighed. After removing the cover, heat shall be applied at 70°C until the sample reaches constant weight. The sample shall then be cooled in a desiccator and weighed.

4.8.5.2 Calculations. The percentage of total solids shall be calculated as follows:

$$\text{Total solids } \% = \frac{\text{wt of residue}}{\text{wt of sample}} \times 100$$

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The cement covered by this specification is intended for use in retreading tires used on military ground vehicles.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. If first article samples are not required (see 3.1and 4.4).
c. If responsibility for inspection shall be other than as specified (see 4.1).

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d. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
e. If inspection conditions shall be other than as specified (see 4.3).
f. Location of first article tests (see 4.4).
g. If number of lot samples shall be other than as specified (see 4.5.1).
h. Selection of applicable level of preservation, packaging and packing (see 5.1).

6.3 Subject term (key word) listing.

Retread cement
Retreading, pneumatic tires
Rubber tread cement
Tire retread cement
Tread cement, rubber

Custodian:
Army - AT

Preparing activity:
Army - AT

(Project 2640-A191)

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
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	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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